Gradient Descent Cheat Sheet

Gradient Descent (BGD, SGD, MBDG):

Update Step:

$\omega_i \leftarrow \omega_i - \eta \frac{\vartheta i}{\vartheta \omega i}$ (figure Q)

- Where the partial derivative $\frac{\vartheta i}{\vartheta \omega i}$ is the *gradient* and η is the **learning rate**
- For **batch gradient descent (BGD)**, this update step represents one epoch (because it represents the entire training set)
- For **stochastic gradient descent (SGD)**, single randomly selected training sample is used to calculate the gradient $\frac{\vartheta i}{\vartheta \omega i}$
- For **mini-batch gradient descent (MBGD)**, batches of datasets (typically hyperparameter of the second power such 32, 64, 128)
    - Items in batch are chosen **at random**
    - Gradient is calculated for all training examples (So, if you set the hyperparameter to 32 you would calculate the gradient for each of the 32 items in the batch)
    - Those gradients are summed together and then **averaged**
    - That averaged gradient is then used in the update rule (Q)

Update With Momentum

- Velocity (v) accumulates gradients over time (t)
- Each weight has its own independent velocity component

$$v_1 = \beta v_{t-1} + (1 - \beta) \frac{\vartheta z}{\vartheta w_{t-1}}$$

- The weight is then updated using the velocity component (Replace the gradient)

$$\omega_i \leftarrow \omega_i - v_t$$

- Momentum allows for faster convergence, reduced oscillations by considering past gradients